
Changes in Body Surface Temperature Play an Underappreciated Role in the Avian Immune Response

Simon Tapper^{1,*}
Joshua K. R. Tabh¹
Glenn J. Tattersall²
Gary Burness³

¹Environmental and Life Sciences Graduate Program, Trent University, Peterborough, Ontario, Canada; ²Department of Biological Sciences, Brock University, St. Catharines, Ontario, Canada; ³Department of Biology, Trent University, Peterborough, Ontario, Canada

Accepted 11/28/2021; Electronically Published 1/28/2022

ABSTRACT

Fever and hypothermia are well-characterized components of systemic inflammation. However, our knowledge of the mechanisms underlying such changes in body temperature is largely limited to rodent models and other mammalian species. In mammals, high dosages of an inflammatory agent (e.g., lipopolysaccharide [LPS]) typically leads to hypothermia (decrease in body temperature below normothermic levels), which is largely driven by a reduction in thermogenesis and not changes in peripheral vasomotion (i.e., changes in blood vessel tone). In birds, however, hypothermia occurs frequently, even at lower dosages, but the thermoeffector mechanisms associated with the response remain unknown. We immune challenged zebra finches (*Taeniopygia guttata*) with LPS, monitored changes in subcutaneous temperature and energy balance (i.e., body mass, food intake), and assessed surface temperatures of and heat loss across the eye region, bill, and legs. We hypothesized that if birds employ thermoregulatory mechanisms similar to those of similarly sized mammals, LPS-injected individuals would reduce subcutaneous body temperature and maintain constant surface temperatures compared with saline-injected individuals. Instead, LPS-injected individuals showed a slight elevation in body temperature, and this response coincided with a reduction in peripheral heat loss, particularly across the legs, as opposed to changes in energy balance. However, we note that our interpretations should be taken with caution owing to small sample sizes within each treatment. We suggest that peripheral vasomotion, allowing for heat retention, is an underappreciated component of the sickness-induced thermoregulatory response of small birds.

Keywords: avian, body temperature, fever, hypothermia, thermography, vasomotion.

Introduction

In vertebrates, the response to infection is typically characterized by a suite of sickness symptoms, including reduced food intake, activity, and changes in body temperature (T_b ; Owen-Ashley and Wingfield 2007). Changes in T_b can take the form of fever (increase in T_b above normothermia), hypothermia (decrease in T_b below normothermia), or a combination of the two (e.g., a biphasic response consisting of initial hypothermia followed by fever; Romanovsky et al. 2005). During sickness, regulation of T_b is coordinated through behavioral and autonomic responses that together result in increased or decreased heat conservation and heat production (Romanovsky et al. 2005). Behavioral responses may include seeking preferred temperatures and/or adopting postures to conserve or reduce heat loss, while autonomic responses may include adjustments in metabolic rate and peripheral vasomotion. The relative contribution of different thermoeffectors (e.g., shivering thermogenesis, changes in peripheral blood flow) on T_b adjustment is known mostly from mammalian studies, which indicate that the dosage of endotoxin is a critical determinant of the resultant effector responses (reviewed in Romanovsky et al. 2005; Garami et al. 2018; Lopes et al. 2021).

In mammals, the febrile response is mediated by increases in both metabolic heat production and peripheral vasoconstriction, with the relative importance of each varying according to ambient temperature (Rudaya et al. 2005; Garami et al. 2018). In contrast, the hypothermic response is mostly mediated by the inhibition of thermogenesis, without changes in peripheral vasomotion (Roth and Blatteis 2014). In rats exposed to the endotoxin lipopolysaccharide (LPS), vasodilation in the tail remains relatively constant as core T_b falls (Romanovsky et al. 1996; Steiner et al. 2005).

In birds, endotoxin exposure commonly elicits hypothermia instead of fever (Owen-Ashley et al. 2006; reviewed in Owen-Ashley and Wingfield 2007; Burness et al. 2010; Lopes et al. 2014; Sköld-Chiriac et al. 2015; Ben-Hamo et al. 2017). There is some evidence that mammals and birds share similar thermoregulatory pathways during the immune response (Dantonio et al. 2016); however, the role of peripheral vasomotion in birds and the

*Corresponding author; email: simontapper@trentu.ca.